Linke Song

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Unik-lif

Beijing, 100190, China

Objective

Interested in computer systems (OS/Virtualization/TEEs). Looking for a system security research internship.

Education

- **University of Chinese Academy of Sciences, Institute of Information Engineering** *Sept.2024 - now*
  PhD of Computer Systems Organization Beijing, China
  Advisor: Prof. Wei Song, Prof. Wenhao Wang
- **University of Chinese Academy of Sciences, Institute of Information Engineering** *Sept.2022 - Jul. 2024*
  Master of Cyberspace Security Beijing, China
  Advisor: Prof. Wenhao Wang
  - GPA: 3.83/4.00
- **University of Chinese Academy of Sciences** *Sept.2018 - Jul. 2022*
  Bachelor of Cyberspace Security Beijing, China
  Advisor: Prof. Wenhao Wang, Prof. Dongdai Lin
  - GPA: 3.70 (6/20)

Research Projects

- **NaCRE: Native confidential Container with RISC-V Enforcement (WIP)** *May. 2025 - now*
  Tools: C, OpenSBI, Linux Kernel, RunC, Qemu code
  - Hardware-software co-design on RISC-V for scalable, native-like confidential containers — avoiding invasive kernel modifications while preserving security with minimal performance overhead.
  - Analyzed memory sharing among container processes; building prototype with customized OpenSBI, Linux Kernel, and RunC.
- **LLM sidechannel** *Jun. 2024 - Apr. 2025*
  Tools: Python code
  - Discovered that KV Cache timing differences in LLM serving systems can leak system prompts; characterized cache hit/miss timing in SGLang with Python scripts.
  - Designed a token-by-token extraction algorithm. **Submission**: Security'25 (RE), CCS'25 (RE), TIFS'25 (AC)
- **NestedSGX** *Jul. 2023 - May. 2024*
  Tools: Rust, C, Python, Linux Kernel Module, Qemu code
  - Designed an Inter-VM security hardening method; implemented a customized Linux-svsm monitor for NestedSGX Enclave with strong performance and compatibility.
  - Developed Guest Kernel Module and automated performance analysis scripts; received 2 AE badges. **Submission**: ASPLOS'24 (RE), CCS'24 (RE), NDSS'25 (AC)

Patents and Publications

C=Conference, J=Journal, P=Patent, S=In Submission, T=Thesis

[J.1] The Early Bird Catches the Leak: Unveiling Timing Side Channels in LLM Serving Systems.
[**Linke Song**, Zixuan Pang], Wenhao Wang*, Zihao Wang, XiaoFeng Wang, Hongbo Chen, Wei Song, Yier Jin, Dan Meng, Rui Houu
TIFS' 25

[C.1] The Road to Trust: Building Enclaves within Confidential VMs.
Wenhao Wang, **Linke Song** (first student author), Benshan Mei, Shuang Liu, Shijun Zhao, Shoumeng Yan*, XiaoFeng Wang, Dan Meng, Rui Hou
NDSS'25

Skills

- **Programming Languages:** C, Java, Rust, Python, lisp
- **DevOps & Version Control:** Git, Docker
- **Specialized Area:** Trusted Execution Environment, Operating System, Virtualization, Container
- **Familiar Architecture:** x86, RISC-V